

P0557

Diagnostic information - DME (DFI) control unit

Brake vacuum sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Brake vacuum sensor faulty

Fault effects

- ♦ Start/Stop system not active

P0558

Diagnostic information - DME (DFI) control unit

Brake vacuum sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Brake vacuum sensor faulty

Fault effects

- ♦ Start/Stop system not active

P0564

Diagnostic information - DME (DFI) control unit

Cruise control operating stalk

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Fault in steering wheel electronics area

Fault effects

- ♦ Cruise control does not work

P0568

Diagnostic information - DME (DFI) control unit

Cruise control operating stalk

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Fault in steering wheel electronics area

Fault effects

- ♦ Cruise control does not work

Diagnostic information - DME (DFI) control unit

Brake light switch

Diagnostic conditions

- Ignition on or engine running
- 10 times in total:
 - ♦ Actuate brake pedal with > 30 bar brake pressure,
 - ♦ then release brake pedal,
 - ♦ then actuate accelerator pedal briefly

Possible fault causes

- ♦ Pressure sensor in the hydraulic unit (PSM) faulty
 - See PSM fault memory
- ♦ Power supply or ground supply faulty (**brake light switch**)
- ♦ Short circuit to B+/ground or open circuit in the line (signal line)
- ♦ **Brake light switch** faulty

Fault setting condition(s)

Fault 1:

- Incorrect value from pressure sensor in the hydraulic unit (PSM)

Fault 2:

- Signal indicating brake actuated and then released again comes from PSM control unit;
- the following is simultaneously detected when the accelerator pedal is actuated after the brake is released:
 - ♦ Brake pedal actuated signal **directly from the brake light switch**,
 - ♦ Brake NOT actuated signal **from the PSM control unit**.

Fault 3:

- The following is detected simultaneously:
 - ♦ Brake pedal NOT actuated signal **directly from the brake light switch**,
 - ♦ Brake actuated signal **from PSM control unit** (with vehicle speed > 0 km/h (0 mph) with > 30 bar brake pressure).

P0597

Diagnostic information - DME (DFI) control unit

Thermostat driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Thermostat faulty

P0598

Diagnostic information - DME (DFI) control unit

Thermostat driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Thermostat faulty

P0599

Diagnostic information - DME (DFI) control unit

Thermostat driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Thermostat faulty

P059F

Diagnostic information - DME (DFI) control unit

Radiator shutter 1 (left): servo motor turns freely or internal fault

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

- ♦ Radiator shutter mechanism broken
- ♦ Servo motor faulty

Fault setting conditions

- No increased current draw when the learned limit stops are reached
- or
- Internal fault, servo motor indicates that it is faulty

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

Diagnostic information - DME (DFI) control unit

Radiator shutter 1 (left): mechanically blocked

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

- ♦ Foreign object stuck between the vanes
- ♦ Radiator shutter mechanism sluggish
- ♦ Servo motor faulty
- For **DTC-DFCC** 00EBCC, 00EBD1 or 00EBD3:
 - ♦ Mechanism blocked with ice

Fault setting conditions

- Increased current draw during movement

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

P05A1

Diagnostic information - DME (DFI) control unit

Radiator shutter 1 (left): perform adaptation

Diagnostic conditions

- Driving at a speed above the initialisation threshold, i.e. the shutter moves to both end positions (20 km/h (12 mph)).

Possible fault causes

- ♦ Previous repairs
- ♦ The DME control unit was re-programmed previously
- ♦ Shutter or adjuster faulty

Fault setting conditions

- Initialization failed

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

Diagnostic information - DME (DFI) control unit

Radiator shutter 1: live code implausible

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply to the radiator shutter
- ♦ Open circuit in ground line for radiator shutter
- ♦ Servo motor faulty

Fault setting condition(s)

The DME control unit expects to receive a signal cyclically on the LIN communication line from the radiator shutter. The fault is set if this signal is not received even though the communication line itself is not faulty.

P05AE

Diagnostic information - DME (DFI) control unit

Radiator shutter 2 (right): servo motor turns freely or internal fault

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

- ♦ Radiator shutter mechanism broken
- ♦ Servo motor faulty

Fault setting conditions

- No increased current draw when the learned limit stops are reached
- or
- Internal fault, servo motor indicates that it is faulty

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

Diagnostic information - DME (DFI) control unit

Radiator shutter 2 (right): mechanically blocked

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

- ♦ Foreign object stuck between the vanes
- ♦ Radiator shutter mechanism sluggish
- ♦ Servo motor faulty
- For **DTC-DFCC** 00EBD6, 00EBDB or 00EBDD:
 - ♦ Mechanism blocked with ice

Fault setting conditions

- Increased current draw during movement

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

P05B0

Diagnostic information - DME (DFI) control unit

Radiator shutter 2 (right): perform adaptation

Diagnostic conditions

- Driving at a speed above the initialisation threshold, i.e. the shutter moves to both end positions (20 km/h (12 mph)).

Possible fault causes

- ♦ Previous repairs
- ♦ The DME control unit was re-programmed previously
- ♦ Shutter or adjuster faulty

Fault setting conditions

- Initialization failed

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

Diagnostic information - DME (DFI) control unit

Radiator shutter 2: live code implausible

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply to the radiator shutter
- ♦ Open circuit in ground line for radiator shutter
- ♦ Servo motor faulty

Fault setting condition(s)

The DME control unit expects to receive a signal cyclically on the LIN communication line from the radiator shutter. The fault is set if this signal is not received even though the communication line itself is not faulty.

P05C0

Diagnostic information - DME (DFI) control unit

Radiator shutter 1 (left): servo motor overtemperature

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

- ♦ Ambient temperature too high
- ♦ Servo motor overloaded
- ♦ Servo motor faulty

Fault setting conditions

- Temperature in servo motor too high

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

P05C1

Diagnostic information - DME (DFI) control unit

Radiator shutter 1 (left): servo motor overtemperature

Diagnostic conditions

- Allow the radiator shutter to move to both learned limit stops (e.g. via drive link)

Possible fault causes

- ♦ Ambient temperature too high
- ♦ Servo motor overloaded
- ♦ Servo motor faulty

Fault setting conditions

- Temperature in servo motor too high

Fault effects

- ♦ New initialization
- ♦ When the fault is active, the DME control unit sends the OPEN request to the radiator shutter servo motor

P0601

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

P0603

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

P0604

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

P0605

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

P0606

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

P0607

Diagnostic information - DME (DFI) control unit

DME control unit in emergency mode

Diagnostic conditions

- Switch on ignition > 10 second(s)
- Then idling > 30 second(s)
- Then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle stationary

Possible fault causes

- ♦ Faults that lead to limp-home mode (see ⓘ)
- ♦ DME (DFI) control unit software error
 - ▶ Re-programming (see ⓘ)

ⓘ INFORMATION

When a serious fault is detected in the engine control system (control unit or parts such as the throttle valve, etc.), the control unit switches to a limp-home program and simultaneously stores a corresponding fault. The **Reduced engine power** warning message is displayed in the instrument cluster.

- ♦ This fault might not be detected again immediately after a restart.
- ♦ Fault P0607 is therefore stored as well, and the control unit already switches to the limp-home program during the restart.
- If one or more other faults are entered in addition to fault P0607:
 - ▶ first correct the other faults,
 - ▶ then erase the fault memory and meet the diagnostic conditions.
- If only fault P0607 is stored, there is an error in the software.
 - ▶ Re-programming DME (DFI) control unit

P0610

Diagnostic information - DME (DFI) control unit

Closed-circuit current measurement in gateway active

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Closed-circuit current measurement in the gateway was activated by the PIWIS Tester

Fault effects

- ♦ The **Check Engine** warning is active in the instrument cluster.
- ♦ There is no engine compartment purge fan run-on (engine compartment overheats).
- ♦ All control units enter sleep mode immediately when the ignition is switched off and as a result, many coding values and adaptations cannot be stored.

P0611

Diagnostic information - DME (DFI) control unit

Control unit faulty

Diagnostic conditions

- Ignition on > 10 second(s)
- then idling > 30 second(s)
- then engine speed > 1,000 rpm > 60 seconds with:
 - Vehicle standstill
- then driving > 60 second(s)

Possible fault causes

- ♦ DME (DFI) control unit faulty

Diagnostic information - DME (DFI) control unit

Control unit function monitoring

or

- For **DTC-DFCC** 00F068:
Starter activation with powertrain closed

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

Diagnostic conditions

- Switch on ignition for more than 10 second(s)
- Then idling for more than 30 second(s)
- Then engine speed of over 1,000 rpm for more than 60 seconds with vehicle stationary
- Then drive with restart by Start Stop system

Possible fault causes

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - ♦ Power supply or ground supply faulty ⇒ See P1559
 - ♦ DME (DFI) control unit software error ► Re-program

or

- For **DTC-DFCC** 00F068:
 - ♦ DME (DFI) control unit faulty

Fault setting condition(s)

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - All calculations for functions in the control unit are monitored by a higher-level unit. A fault is entered if monitoring detects implausible calculations.

or

- For **DTC-DFCC** 00F068:
 - The starter was actuated by the Start Stop system although a closed powertrain was detected (gear engaged and clutch closed).

Fault effects

- For **DTC-DFCC** 00F068:
 - ♦ Start Stop system deactivated

Diagnostic information - DME (DFI) control unit

Control unit function monitoring

or

- For **DTC-DFCC** 00F068:
Starter activation with powertrain closed

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

Diagnostic conditions

- Switch on ignition for more than 10 second(s)
- Then idling for more than 30 second(s)
- Then engine speed of over 1,000 rpm for more than 60 seconds with vehicle stationary
- Then drive with restart by Start Stop system

Possible fault causes

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - ♦ Power supply or ground supply faulty ⇒ See P1559
 - ♦ DME (DFI) control unit software error ► Re-program

or

- For **DTC-DFCC** 00F068:
 - ♦ DME (DFI) control unit faulty

Fault setting condition(s)

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - All calculations for functions in the control unit are monitored by a higher-level unit. A fault is entered if monitoring detects implausible calculations.

or

- For **DTC-DFCC** 00F068:
 - The starter was actuated by the Start Stop system although a closed powertrain was detected (gear engaged and clutch closed).

Fault effects

- For **DTC-DFCC** 00F068:
 - ♦ Start Stop system deactivated

P0627

Diagnostic information - DME (DFI) control unit

Fuel pump relay final stage

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel pump relay faulty

P0628

Diagnostic information - DME (DFI) control unit

Fuel pump relay final stage

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel pump relay faulty

P0629

Diagnostic information - DME (DFI) control unit

Fuel pump relay final stage

Diagnostic conditions

- Engine running > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel pump relay faulty

P0638

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit

Diagnostic conditions

- Vehicle standstill
- Coolant temperature 5 ... 100 °C (41 ... 212 °F)
- Intake air temperature 5 ... 30 °C (41 ... 86 °F)
- Accelerator pedal not pressed (0%)
- Ignition off
- then switch on ignition > 60 second(s) with engine off

Possible fault causes

- ♦ Adaptation was not performed (successfully).
- ♦ Throttle valve adjusting unit faulty

P0642

Diagnostic information - DME (DFI) control unit

Sensor voltage 1

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 62:

- ♦ Tank vent pressure sensor
- ♦ Oil pressure sensor

Connector [A], contact 77:

- ♦ Throttle valve adjusting unit

Connector [B], contact 59:

- ♦ Clutch rpm sensor faulty (manual transmission)

Connector [B], contact 72

- ♦ Pedal sensor 1 faulty

Connector [B], contact 75:

- ♦ Neutral gear sensor faulty (manual transmission)

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0642)
 - or
 - ♦ > 5.3 V (P0643)

P0643

Diagnostic information - DME (DFI) control unit

Sensor voltage 1

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 62:

- ♦ Tank vent pressure sensor
- ♦ Oil pressure sensor

Connector [A], contact 77:

- ♦ Throttle valve adjusting unit

Connector [B], contact 59:

- ♦ Clutch rpm sensor faulty (manual transmission)

Connector [B], contact 72

- ♦ Pedal sensor 1 faulty

Connector [B], contact 75:

- ♦ Neutral gear sensor faulty (manual transmission)

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0642)
 - or
 - ♦ > 5.3 V (P0643)

P0652

Diagnostic information - DME (DFI) control unit

Sensor voltage 2

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 37:

- ♦ Oil pressure sensor

Connector [A], contact 80:

- ♦ Intake manifold pressure sensor
- ♦ Boost pressure sensor
- ♦ Exhaust temperature sensor, bank 1
- ♦ Exhaust temperature sensor, bank 2

Connector [A], contact 81:

- ♦ Fuel high-pressure sensor
- ♦ Fuel low-pressure sensor

Connector [B], contact 07:

- ♦ Vacuum sensor

Connector [B], contact 12:

- ♦ Pedal sensor 2

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0652)
 - or
 - ♦ > 5.3 V (P0653)

P0653

Diagnostic information - DME (DFI) control unit

Sensor voltage 2

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 37:

- ♦ Oil pressure sensor

Connector [A], contact 80:

- ♦ Intake manifold pressure sensor
- ♦ Boost pressure sensor
- ♦ Exhaust temperature sensor, bank 1
- ♦ Exhaust temperature sensor, bank 2

Connector [A], contact 81:

- ♦ Fuel high-pressure sensor
- ♦ Fuel low-pressure sensor

Connector [B], contact 07:

- ♦ Vacuum sensor

Connector [B], contact 12:

- ♦ Pedal sensor 2

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0652)
 - or
 - ♦ > 5.3 V (P0653)

P0658

Diagnostic information - DME (DFI) control unit

Power supply, terminal 15 or 87

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Vehicle electrical system supply voltage too high (e.g. jump-lead starting)
- ♦ Vehicle electrical system supply voltage too low
- ♦ Contact resistance, loose contact or corrosion on connectors or lines (DME (DFI) control unit)

Fault setting condition(s)

- Deviation in supply voltage at terminal 15 or 87:

- ♦ < 10 V (P0658/P2531)

or

- ♦ > 16 V (P0659/P2532)

at:

- ♦ DME (DFI) control unit - connector [B], contact 50 ⇒ Terminal 15
 - ♦ DME (DFI) control unit - connector [B], contact 3/5/6 ⇒ Terminal 87

P0659

Diagnostic information - DME (DFI) control unit

Power supply, terminal 15 or 87

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Vehicle electrical system supply voltage too high (e.g. jump-lead starting)
- ♦ Vehicle electrical system supply voltage too low
- ♦ Contact resistance, loose contact or corrosion on connectors or lines (DME (DFI) control unit)

Fault setting condition(s)

- Deviation in supply voltage at terminal 15 or 87:
 - ♦ < 10 V (P0658/P2531)or
 - ♦ > 16 V (P0659/P2532)
- at:
- ♦ DME (DFI) control unit - connector [B], contact 50 ⇒ Terminal 15
 - ♦ DME (DFI) control unit - connector [B], contact 3/5/6 ⇒ Terminal 87

P0685

Diagnostic information - DME (DFI) control unit

Start delay due to undervoltage

Diagnostic conditions

- DME (DFI) control unit is viable (voltage > 5 V at B50 (term.15) + ground)
- Ignition is switched on (voltage present at B50)

Possible fault causes

- If the actual value > **A010-DME SUPPLY VOLTAGE** < in the FreezeFrame of the fault entry P0685 was stored at < 8.0 V:

- ♦ Voltage is too low or voltage will drop when the engine starts (see ⓘ).

- If the actual value > **A010-DME SUPPLY VOLTAGE** < in the FreezeFrame of the fault entry P0685 was stored at > 8.0 V:

- ♦ Open circuit in power supply for main relay control circuit
- ♦ Short circuit to ground or open circuit in the lines between the main relay and DME (DFI) control unit (DME (DFI) control unit power supply)
- ♦ Short circuit to B+ or open circuit in the lines between the DME (DFI) control unit and main relay (main relay activation)
- ♦ Main relay faulty

Fault setting condition(s)

- A check is carried out once to determine whether the voltage at B3/B5/B6 has exceeded the min. value of 5.0 V.
- Test duration 0.0 to approx. 0.5 seconds after ignition on, or after control unit reset.

Fault effects

- ♦ Fuel injection is blocked.

ⓘ INFORMATION

- ♦ According to specifications, the DME main relay must work only at 8.0 V or higher.
- ♦ There are also relays that work at a significantly lower voltage.
- ♦ The entire engine control system can work at 5.0 V or higher.
- ♦ If the start voltage drops, this can cause the diagnosis to detect the fault > **START DELAY DUE TO UNDERVOLTAGE** < and fuel injection will be blocked.
- ♦ The starter relay can also pick up at 3 V.
- ♦ This means that the starter might still turn, but the engine cannot start.

Possible behavior and displays:

- Start voltage < 5 V:
 - ♦ Starter might turn
 - ♦ Engine does not start
 - ♦ NO DME fault entries because the DME (DFI) control unit is not viable.
- Start voltage > 5 V ... < 8 V

- + DME main relay does NOT work:
 - ♦ Starter turns
 - ♦ Engine does not start
 - ♦ DME fault entry P0685 with FreezeFrame actual value
- > **A010-DME SUPPLY VOLTAGE** < = 0 V
 - ♦ NO DME fault entry P0686 because these diagnostic conditions (> 8.0 V at A34) were not met.
- Start voltage > 5 V ... < 8 V
 - + DME main relay is working:
 - ♦ Starter turns
 - ♦ Engine starts
 - ♦ NO DME fault entries
- Start voltage > 8 V
 - + DME main relay does NOT work:
 - ♦ Starter turns
 - ♦ Engine does not start
 - ♦ DME fault entry P0685 with FreezeFrame value
 - > **A010-DME SUPPLY VOLTAGE** < = 0 V
 - ♦ DME fault entry P0686 with FreezeFrame value
 - > **A010-DME SUPPLY VOLTAGE** < = 0 V
- Start voltage > 8 V
 - + DME main relay is working:
 - ♦ Starter turns
 - ♦ Engine starts
 - ♦ NO DME fault entries